

CSCI 264 — Assembly Language

Lab 4

Strings and Structs in C

Start from the program we wrote together in class on Friday (on the course website as `lecture_10.c`). Copy it into your directory for this course and rename it `roster.c`. It already defines `struct studentT`, allocates an array of five students on the heap, fills them in with `init_student`, and prints them.

You will add two functions and a few lines to `main`. Do not change the struct definition or `init_student`.

Functions

Write each of these functions:

```
struct studentT *find(struct studentT *roster, int n, char *name);
void print_offset(struct studentT *s);
```

- `find` searches the roster for the student whose `name` field matches `name`. It returns a pointer to that student, or `NULL` if nobody on the roster has that name.
- `print_offset` takes a pointer to one student struct and prints the address of the struct itself, then the address of each of its four fields and that field's *offset* — the number of bytes from the start of the struct to the start of the field.

Offsets

An offset is a subtraction of memory addresses: the address of the field minus the address of the struct. Cast both to `long` first, so that you are subtracting two ordinary numbers:

```
(long)&s->age - (long)s
```

This is the same subtraction you would do by hand with the two hexadecimal addresses your program just printed. Print the result for each field like this:

```
printf("[offset] s          = %p\n", s);
printf("[offset] &s->name    = %p   offset %ld\n",
       s->name, (long)s->name - (long)s);
```

Add to main

After the loop that prints the roster:

1. Call `find` twice — once with a name that is on the roster, and once with a name that is not. Print the student you found in the first case and a message in the second.
2. Call `print_offset` on the student that `find` returned.
3. Print `sizeof(struct studentT)` using `%lu`.
4. Print `roster`, `roster + 1`, and `roster + 2`.

Output

A run of your program should look like this, with real addresses in place of `0x...`:

```
... the five students, printed by the loop already in the program ...
```

```
found Falilou    age 20  gpa 3.5  class of 2028
Ada Lovelace is not on the roster
```

```
[offset] s          = 0x...
[offset] &s->name    = 0x...  offset 0
[offset] &s->age      = 0x...  offset 64
[offset] &s->gpa      = 0x...  offset 68
[offset] &s->grad_yr  = 0x...  offset 72
```

```
[main]  sizeof(struct studentT) = 76
[main]  roster      = 0x...
[main]  roster + 1 = 0x...
[main]  roster + 2 = 0x...
```

Submission

Submit `roster.c` on Canvas and demo the program before you leave lab. If you do not finish during the period, submit on Canvas by Tuesday, September 29 and demo during office hours. Put your name in a comment at the top of the source file. If a classmate helped you put their name in the comment. Disclose any use of Generative AI in a comment at the top.